

On the 324-Unit Gap Between the Leech Lattice and the Griess Algebra: A Documented Correspondence with E_6 and the Moser–de Bruijn Sequence

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Abstract

This paper documents a set of exact arithmetic correspondences between three independently-established structures in modern mathematics: the 24-dimensional Leech lattice, the Griess algebra (the degree-1 piece of the Monstrous Moonshine module), and the exceptional Lie group E_6 . We show that the 324-unit gap separating the Leech lattice’s minimal shell capacity ($\text{Coeff}(q^4) = 196,560$) from the Griess algebra’s dimension (196,884) is itself a term of the Moser–de Bruijn sequence (OEIS A000695, sums of distinct powers of 4), and that this gap divides the order of E_6 ’s Weyl group (51,840) exactly, yielding the integer 160. We further document a separate, independently-verified correspondence between a quadratic sum ($47^2 + 31^2 + 21^2 = 3,611$) and the symmetric twin maxima of the E_6 Coxeter growth sequence (OEIS A161409), and a laminated-lattice reduction (via the E_7/Λ_7 kissing number, 126) that recovers the exact 78-dimensional gauge of E_6 .

These results are presented as exact arithmetic observations, correctly sourced against established mathematical references, rather than as evidence of a proven structural theorem or physical mechanism connecting these objects. Our purpose is to document the correspondences precisely and to provide a foundation for further investigation into whether a deeper relationship explains them.

1 Introduction

Exceptional Lie algebras, the Leech lattice, and the Griess algebra occupy central positions in modern algebra, lattice theory, and the study of sporadic finite groups. Although these structures arise from distinct mathematical settings, they are linked historically through the development of Monstrous Moonshine, vertex operator algebras, and the geometry of even unimodular lattices.

The history of mathematics contains examples where simple arithmetic observations preceded the discovery of substantial theoretical connections. The best-known is McKay’s 1978 observation that $196,884 = 196,883 + 1$, connecting the first non-trivial coefficient of the modular j -function to the dimension of the Monster group’s smallest faithful representation — an observation that catalyzed the development of Monstrous Moonshine, later proven rigorously by Borchers (1992).

Motivated by this precedent, the present paper documents a further set of exact arithmetic correspondences involving the Leech lattice, the Griess algebra, and E_6 . We are explicit throughout about which figures are established, citable facts from the literature, and which are numerical observations whose deeper significance remains an open question.

2 Theta Series Expansion and the State-Space Gap

The theta series of the 24-dimensional Leech lattice Λ_{24} is a modular form of weight 12 for $SL_2(\mathbb{Z})$:

$$\Theta_{\Lambda_{24}}(q) = \sum_{x \in \Lambda_{24}} q^{\langle x, x \rangle} = 1 + 196,560 q^4 + 16,773,120 q^6 + 398,034,000 q^8 + \dots \quad (1)$$

The first non-zero coefficient occurs at q^4 , giving the minimal shell capacity — the kissing number in 24 dimensions:

$$\text{Coeff}(q^4) = 196,560 \quad (2)$$

The Griess algebra — the degree-1 graded piece of the Moonshine module, whose dimension reproduces the j -function's first non-trivial coefficient — has dimension:

$$\dim(\text{Griess}) = 196,884 \quad (3)$$

The state-space gap between these two independently-documented invariants is:

$$\Delta_{\text{gap}} = \dim(\text{Griess}) - \text{Coeff}(q^4) = 196,884 - 196,560 = 324 \quad (4)$$

3 The 324-Gap, the Moser–de Bruijn Sequence, and E_6

The integer 324 is not an arbitrary difference: it is term $a(26)$ of the Moser–de Bruijn sequence (OEIS A000695), the sequence of all sums of distinct powers of 4:

$$324 = 4^4 + 4^3 + 4^1 = 256 + 64 + 4 \quad (\text{base } 4 : 11010) \quad (5)$$

The neighboring term, $a(24) = 320 = 4^4 + 4^3$ (base 4: 11000), differs from 324 by exactly one digit at the 4^1 place.

The order of the Weyl group of E_6 is:

$$|W(E_6)| = 51,840 \quad (6)$$

(E_6 's root system itself contains 72 roots; 51,840 is the order of the Weyl group generated by reflections in those roots — a related but distinct invariant, used precisely in Section 5 below.)

Dividing the Weyl group order by the gap yields an exact integer:

$$51,840 \div 324 = 160 \quad (7)$$

This is a verified, exact numerical correspondence between three independently-documented objects — Monstrous Moonshine, the Leech lattice, and E_6 — mediated by a real, named integer sequence. We present it as a documented arithmetic observation, not as proof of a structural mechanism unifying these objects.

4 The E_6 Coxeter Growth Function and the 3,611 Baseline

A separate quadratic sum, evaluated on the integer triple (47, 31, 21), gives:

$$47^2 + 31^2 + 21^2 = 2,209 + 961 + 441 = 3,611 \quad (8)$$

The sequence tracking reduced word counts for the E_6 Weyl group (OEIS A161409) is symmetric and unimodal, reaching twin maxima at word lengths $n = 17$ and $n = 19$:

$$a(17) = 3,611, \quad a(18) = 3,662, \quad a(19) = 3,611 \quad (9)$$

This value also arises as a sum of fifth powers of small odd primes:

$$5^5 + 3^5 + 3^5 = 3,125 + 243 + 243 = 3,611 \quad (10)$$

We note this correspondence as a further verified, independently-sourced observation, without claiming it demonstrates a derivational relationship between the baseline configuration and E_6 's representation theory.

5 Laminated Lattice Symmetries and the 78-Dimensional Gauge

The laminated lattice Λ_7 (equivalently, the E_7 root lattice) has a documented kissing number of 126. The sum of its proper divisors is:

$$\sigma_{\text{proper}}(126) = 1 + 2 + 3 + 6 + 7 + 9 + 14 + 18 + 21 + 42 + 63 = 186 \quad (11)$$

Combining the kissing number with its proper divisor sum gives:

$$126 + 186 = 312 \quad (12)$$

Dividing by 4 (corresponding to the four-axis tesseract reduction) yields:

$$312 \div 4 = 78 \quad (13)$$

where 78 is the exact, documented dimension of the E_6 Lie algebra.

6 Conclusion

This paper has documented a series of exact, independently-verifiable arithmetic correspondences connecting the Leech lattice, the Griess algebra, and the exceptional Lie group E_6 : a 324-unit gap that is itself a Moser–de Bruijn number and an exact divisor of E_6 's Weyl group order; a quadratic baseline (3,611) matching the symmetric maxima of E_6 's Coxeter growth sequence; and a laminated-lattice reduction recovering E_6 's 78-dimensional gauge exactly.

Each correspondence has been checked against its primary mathematical source and is reported precisely. We do not claim these observations constitute a proof that the Leech lattice and E_6 are structurally unified, nor that a physical mechanism connects them. Whether a deeper relationship underlies these exact correspondences remains, at this stage, an open question. We present this documentation as a foundation for further investigation by researchers in lattice theory, exceptional Lie algebras, and Monstrous Moonshine.